KAVEESHA DHANANJAYA

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https://kd-27.github.io/portfolio/

RESEARCH INTERESTS

Legged Robotics and Locomotion Control; Motion Planning and Trajectory Optimization; Swarm Robotics and Multi-Agent Coordination; Topological Methods in Path Planning; SLAM and Autonomous Navigation; Computer Vision for Robotic Perception.

EDUCATION

BSc Engineering (Hons) in Mechatronics Engineering

General Sir John Kotelawala Defence University, Sri Lanka 2021 – 2025

Cumulative GPA: 3.91 / 4.00 — **Gold Medalist, Best Academic Performance**

Relevant Coursework: Advanced Robotics, Machine Elements in Design, Power Electronics, Digital Signal Processing, Control Systems, Machine Learning

PUBLICATIONS

[1] GK. Dhananjaya et al., "**Design and Development of a Quadruped Robot for Inspection in Human-Inaccessible Structured Areas**" *Under review, ICROS-KIEE International Conference*, 2025.

Developed a novel quadruped robot employing five-bar parallel leg mechanisms. Introduced a geometric Variable Circle Method for inverse kinematics. Achieved 20 cm/s locomotion, 30°/s rotation and 120–130 m wireless range via ESP-NOW. Implemented ROS2-based distributed control with MicroROS integration.

[2] GK. Dhananjaya et al., "**Prediction of Coronary Artery Disease using Artificial Neural Network**" *KDU 17th International Research Conference*, 2024. (Oral Presentation)

Designed an ANN achieving ~90% diagnostic accuracy for CAD. Applied SMOTE for class imbalance, collaborated with cardiologists for clinically-informed feature selection, and employed stratified k-fold cross-validation with dropout and batch normalization.

[3] Co-author, "**Development of an Automated Clothesline System**" *KDU 16th International Research Conference*, 2023. (Poster Presentation)

Designed and integrated sensor-actuator system for autonomous clothes drying with weather detection, light sensing, and remote control capabilities.

RESEARCH EXPERIENCE

Quadruped Robot for Inspection in Human-Inaccessible Environments 2024 – 2025

Final Year Research Project | Supervisor: Mr. Isira Naotunna

- Designed five-bar parallel leg mechanism optimizing for efficiency and simplicity over conventional serial architectures
- Developed geometric Variable Circle Method for inverse kinematics, reducing computational complexity while maintaining positioning accuracy
- Conducted MATLAB simulations for trajectory optimization, achieving stability score of 987.9
- Implemented ROS2 communication architecture with MicroROS for distributed real-time control

Wheeled Robot with LiDAR-Based SLAM for Mapping and Navigation 2023

Third Year Research Project | Supervisor: Mr. Isira Naotunna

- Designed and fabricated Mars Rover-inspired wheeled robot equipped with LiDAR for autonomous navigation
- Implemented SLAM algorithm in ROS achieving repeatable localization accuracy of ±10 cm position and ±15° orientation
- Developed kinematic model and conducted GAZEBO simulation for system validation prior to deployment

INDUSTRY EXPERIENCE

Senior Robotics Engineer September 2025 – Present

Hype Insight (Pty) Ltd

- Design and development of autonomous wheeled robot integrating AI-driven computer vision systems
- Conduct R&D on rover and drone platforms for specialized applications
- Developing microphone array system with custom audio processing pipeline

Product Development Engineer – Innovations April 2025 – August 2025

MAS Capital (Pvt) Ltd

- Conducted feasibility analysis and developed mechanical systems for process efficiency improvements

Automation Engineering Intern October 2023 – March 2024

MAS Capital (Pvt) Ltd – Automation Lab

- Designed and fabricated three PLC-controlled modular units for garment manufacturing automation
- Developed CNN for pick-and-place orientation detection achieving 85% accuracy; conceptualized UR5 integration

HONORS & AWARDS

Gold Medal – Best Academic Performance, Mechatronics Engineering, KDU 2025

1st Place & Best Design Award, KDU Mechatronics Systems Design Competition 2023

President Scout Award, Royal College Colombo 2022

TECHNICAL SKILLS

Robotics & Control: ROS, GAZEBO, MATLAB/Simulink/Simscape, Inverse Kinematics, Motion Planning

Machine Learning: PyTorch, TensorFlow, Scikit-learn, OpenCV, YOLO

Programming: Python, C++

CAD & Simulation: SolidWorks, Fusion 360, ANSYS

Electronics: PLC Programming (Siemens, Delta), PCB Design (KiCAD, EasyEDA), Proteus, LTSpice

Hardware: Nvidia Jetson, Raspberry Pi, ESP32, Arduino, LiDAR sensors, Depth cameras

Fabrication: 3D Printing (FDM), Laser Cutting, Lathe Machining, Milling

LEADERSHIP & SERVICE

Research In-Charge, Electronic Robotics and Innovation Club, KDU 2022–2023

STEM Educator, 360 Labs – Designed robotics curriculum and led workshops for students 2023–2024

Senior Prefect, Royal College Colombo 2020–2021

REFERENCES

Mr. Isira Naotunna — Senior Lecturer, Department of Mechanical Engineering, KDU

BSc Hons (Moratuwa), M.Eng (CMU Thailand) | Email: isira.naotunna@adelaide.edu.au | Phone: +94 71 627 8136

Relationship: Undergraduate Research Supervisor & Robotics Lecturer

Mr. H M D L Dabare — Senior Automation Engineer, MAS Capital (Pvt) Ltd

Email: mirand@masholdings.com | Phone: +94 77 406 1725

Relationship: Industry Supervisor during Automation Internship